HUMPHREY VIRTUAL FARMS LLC // WP-2: 36.75NS SIC MOSFET INTERLOCK BOARD SPECIFICATION
Entity: Humphrey Virtual Farms LLC (CAGE: 1AHA8 | SAM.gov UEI: S1M4ENLHTDH5)
Controlling Authority: Jeffery Humphrey, Founder, CEO, & Principal Sovereign Architect
Hardware Model: HVF-SIC-3675-REV1 (WP-2 Hardware Interlock) | Enclave Tier: PUBLIC
Host Ebony Baseline Merkle Root:
cc1ee1c8d29b2ae5e856fc0fcb672a52e7e49d50369499febe14e18e9150949a

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1. Deterministic 3-Stage Propagation Timing Budget (Total = 36.75 Nanoseconds)
- Stage 1 (Analog Window Comparator): 18.50 ns
- Stage 2 (5kVrms Isolated Gate Driver): 12.00 ns
- Stage 3 (1200V SiC MOSFET Turn-Off): 6.25 ns
- TOTAL HARDWARE GUILLOTINE LATENCY: 36.75 ns (4,081,632.7x faster than legacy 150ms PLC)

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2. Physical Stopping Kinematics & Inductive Snubber Telemetry
- Tested Manipulator Velocity: 1,500.0 mm/s
- HVF 36.75ns Stopping Distance: 0.055125 microns (um) (0.000055 mm)
- Legacy 150ms PLC Over-Travel Distance: 225.00 mm (Catastrophic crush hazard eliminated)
- DC Actuator Bus Rating: 480.0 V DC @ 85.0 A Peak Fault Current
- Inductive Snubber Clamp Energy (0.5 * L * I^2): 0.043350 Joules
- Peak SiC Junction Temperature: 26.82 C (Limit: < 175.0 C | Pass: True)

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3. Bill of Materials (BOM) & Calibration Network
- Unit PCB Assembly Cost: [REDACTED_PRIVATE_MANUFACTURING_COST_DFARS_252_227_7018]
- Calibration Divider Network: [REDACTED_PROPRIETARY_TRIM_DIVIDER_DFARS_252_227_7018]
- Stage 1: High-Speed Window Comparator (18.50 ns): Dual Ultra-Fast LVDS Comparator (t_pd = 18.50ns max) -- Monitors torque/current shunt & optical curtain at 0.0A neural isolation
- Stage 2: Galvanically Isolated Gate Driver (12.00 ns): 5kVrms Capacitive Isolated SiC Gate Driver (t_pd = 12.00ns) -- Drives -5V/+20V Miller-clamped turn-off across high-side DC actuator bus
- Stage 3: Silicon-Carbide (SiC) Power MOSFET Array (6.25 ns): 1200V / 16mOhm SiC Trench MOSFET Half-Bridge (t_off = 6.25ns) -- Severs 480V DC bus in 6.25ns; TVS array absorbs 0.0433 J

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WORM Hardware Spec Seal (SHA-256):
91ee522311d514c0f906e08e95f834486f6ac4f337db9f2694260fedf36590ab